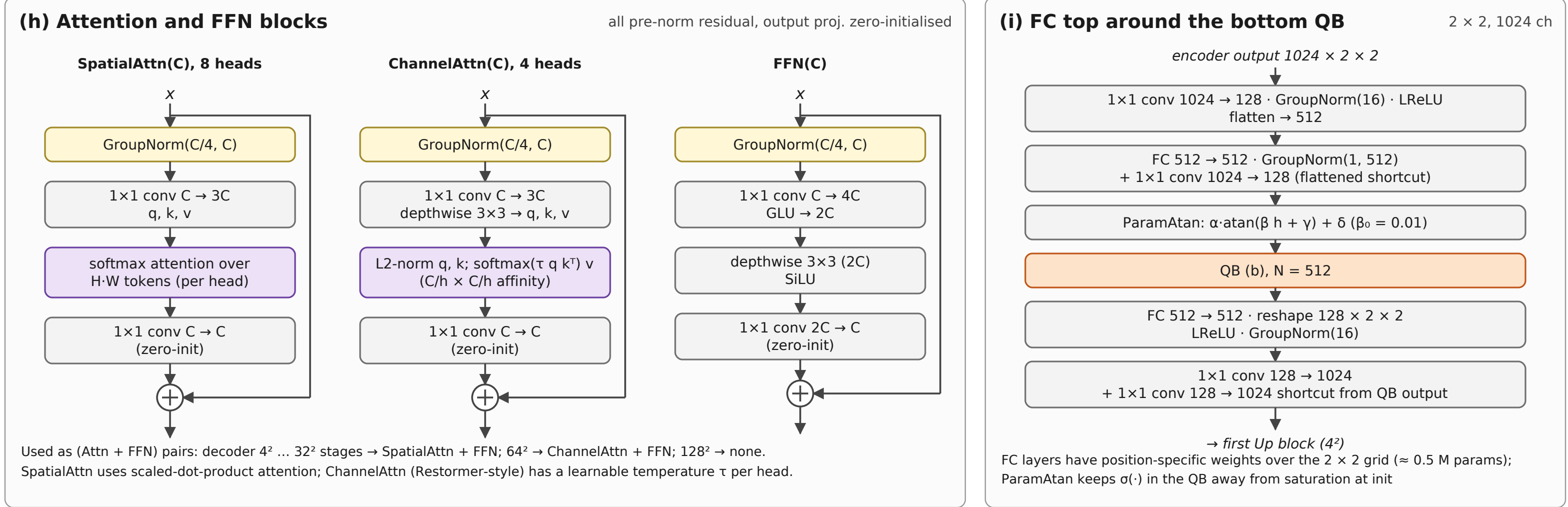
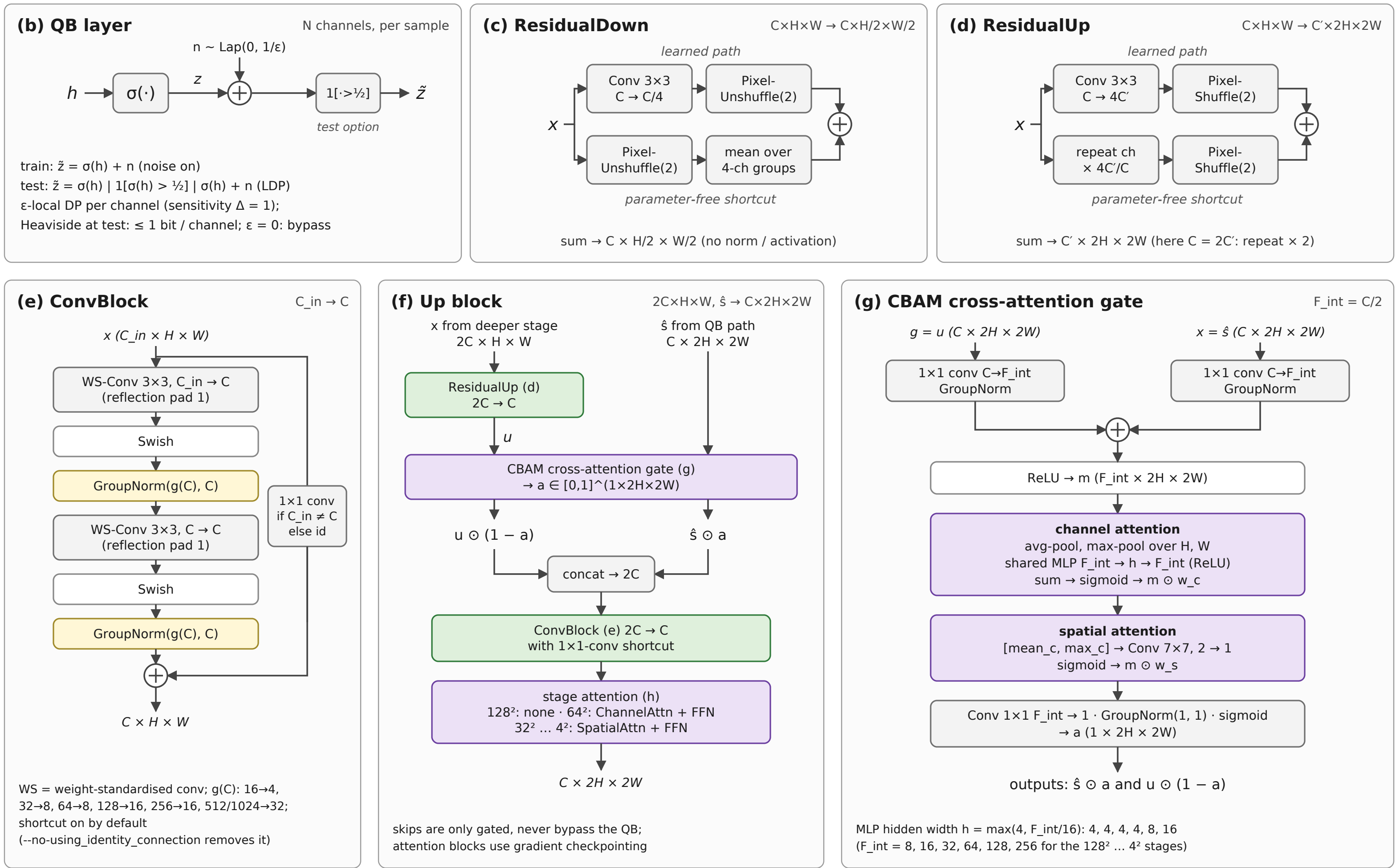
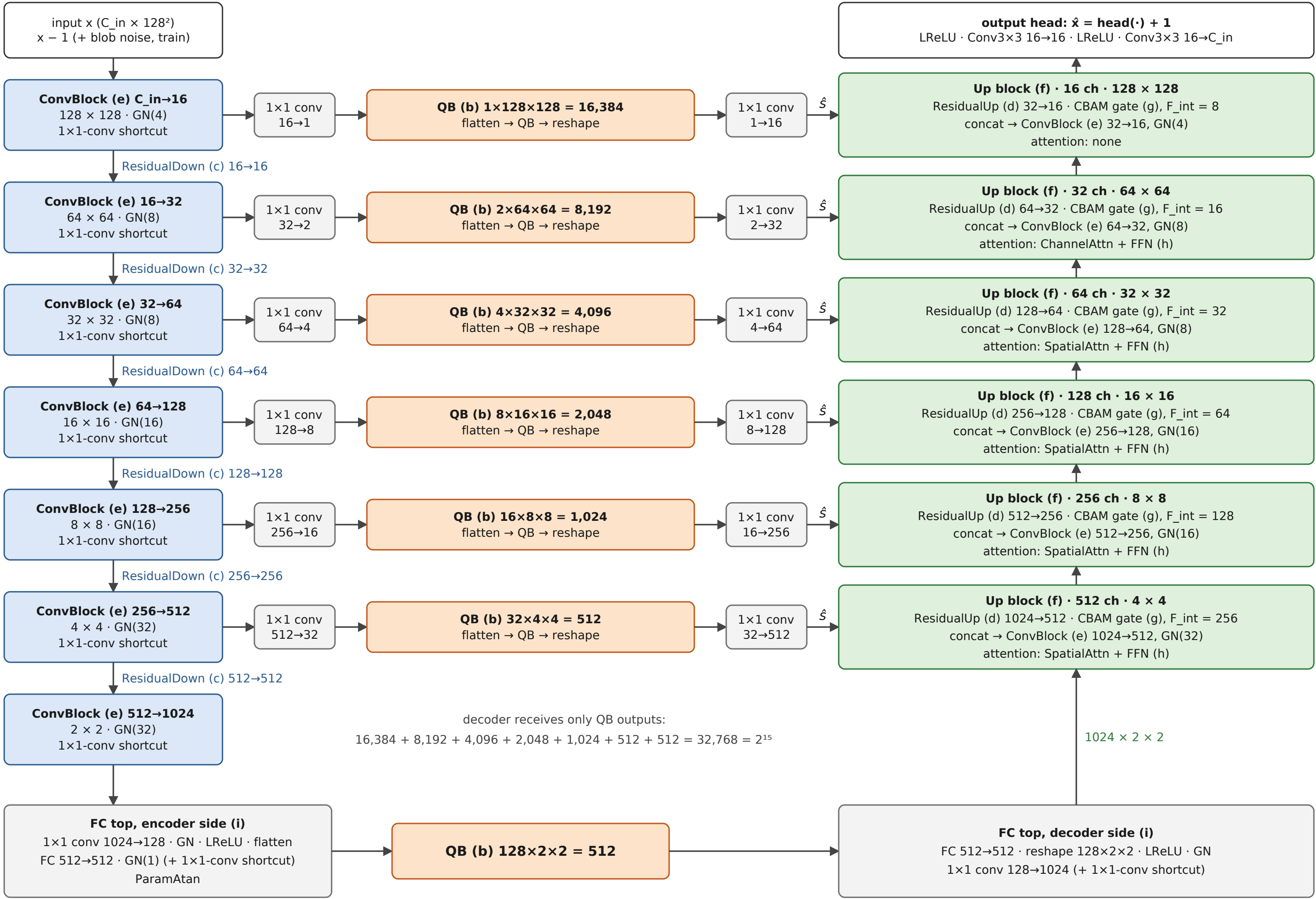


(a) Overall architecture (wf = 4, depth = 7, 128 × 128 input, 60.7 M parameters)



(j) Training objective and anomaly score

train: AdamW, lr $1e-3$, cosine decay to $1e-5$ after 5 warm-up epochs, gradient clipping 1.0; input x + blob noise; all QB layers add Laplace noise; $L = \text{mean}(x - \hat{x})^2 + \lambda_p \cdot L_{\text{perc}}(x, \hat{x})$ [+ optional KL sparsity on $\sigma(h)$]
 L_{perc} : relative L1 between VGG19 relu4_2 features (ImageNet weights), random shift $\leq 8 \text{ px}$ in training; variance head unused (--not_use_log_var)
test: QB noise off (identity), Heaviside or LDP mode; anomaly map = $(x - \hat{x})^2 + \lambda_p \cdot \text{upsampled VGG map}$; image score = perceptual term (AUC_perceptual)